

Linear Algebra I — Week 3

Span, Linear Independence & Setting Up Linear Systems

Name: _____ Date: _____ All work **by hand** — no calculator, no software.

Key Facts

Linear combination and span. A linear combination of $v_1, \dots, v_k \in V$ is $a_1v_1 + \dots + a_kv_k$ with $a_i \in \mathbb{F}$. $\text{span}\{v_1, \dots, v_k\}$ is the set of all of them; $\text{span} \emptyset = \{\mathbf{0}\}$. It is a subspace, and the smallest subspace containing $v_1, \dots, v_k$.

Setting up the system. To test $w \in \text{span}\{v_1, \dots, v_k\}$ in $\mathbb{F}^n$, solve $a_1v_1 + \dots + a_kv_k = w$. Coordinatewise this is n equations in k unknowns, with augmented matrix whose columns are $v_1, \dots, v_k \mid w$. Solution exists $\iff w \in \text{span}$.

Linear independence. $v_1, \dots, v_k$ are *independent* if $a_1v_1 + \dots + a_kv_k = \mathbf{0}$ forces $a_1 = \dots = a_k = 0$; otherwise *dependent*. Test: solve the homogeneous system; only $\mathbf{0} \Rightarrow$ independent, a nonzero solution $\Rightarrow$ dependent (and it gives a relation).

Useful facts. A set containing $\mathbf{0}$ is dependent. One vector is independent iff it is nonzero. Two vectors are independent iff neither is a scalar multiple of the other. $v_1, \dots, v_k$ dependent $\iff$ some $v_j \in \text{span}$ of the others. If $w \in \text{span}\{v_1, \dots, v_k\}$, adding w does not enlarge the span. Independence and span depend on the field: $\{(1, i), (i, -1)\}$ is dependent over $\mathbb{C}$ but independent over $\mathbb{R}$.

Counting. More homogeneous unknowns than equations forces a nonzero solution; so more than n vectors in $\mathbb{F}^n$ are dependent.

Worked Examples

Example 1 · Membership in a span

Is $(4, 7) \in \text{span}\{(1, 2), (3, 5)\}$ in $\mathbb{R}^2$?

Solution.

1. We need numbers a, b with $a(1, 2) + b(3, 5) = (4, 7)$.
2. Expand: $a(1, 2) = (a, 2a)$ and $b(3, 5) = (3b, 5b)$. Add: $(a + 3b, 2a + 5b)$.
3. Match coordinates: Eq. 1: $a + 3b = 4$; Eq. 2: $2a + 5b = 7$.
4. From Eq. 1: $a = 4 - 3b$.
5. Substitute into Eq. 2: $2(4 - 3b) + 5b = 7$, i.e. $8 - 6b + 5b = 7$, i.e. $8 - b = 7$, so $b = 1$.
6. Then $a = 4 - 3(1) = 1$.
7. Check: $1(1, 2) + 1(3, 5) = (1, 2) + (3, 5) = (4, 7) \checkmark$.

Answer: Yes: $(4, 7) = 1(1, 2) + 1(3, 5)$.

Example 2 · Membership (positive)

Is $(1, 2, 3) \in \text{span}\{(1, 0, 1), (0, 1, 1)\}$?

Solution.

1. We seek a, b with $a(1, 0, 1) + b(0, 1, 1) = (1, 2, 3)$.
2. Expand: $a(1, 0, 1) = (a, 0, a)$ and $b(0, 1, 1) = (0, b, b)$. Add: $(a, b, a + b)$.
3. Match coordinates: $a = 1, b = 2, a + b = 3$.
4. The first two equations give the values $a = 1$ and $b = 2$ directly.
5. There are three equations for two unknowns, so check the third: $a + b = 1 + 2 = 3$, which equals the required 3 $\checkmark$.
6. All three equations hold, so the vector is in the span.

Answer: Yes: $(1, 2, 3) = 1(1, 0, 1) + 2(0, 1, 1)$.

Example 3 · Membership (negative)

Is $(1, 2, 4) \in \text{span}\{(1, 0, 1), (0, 1, 1)\}$?

Solution.

1. As in the previous example, $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$.
2. Matching $(1, 2, 4)$ requires $a = 1, b = 2, a + b = 4$.
3. The first two equations force $a = 1$ and $b = 2$.
4. Test the third: $a + b = 1 + 2 = 3$, but the equation requires 4. Since $3 \neq 4$, the system is inconsistent: no a, b satisfy all three equations.
5. So $(1, 2, 4)$ is not a linear combination of the two vectors.

Answer: No.

Example 4 · Dependence, finding a relation

Test $\{(1, 2, 1), (2, 1, 0), (1, -1, -1)\}$ for independence.

Solution.

1. Suppose $a(1, 2, 1) + b(2, 1, 0) + c(1, -1, -1) = \mathbf{0}$.
2. Expand coordinatewise: $a(1, 2, 1) = (a, 2a, a)$, $b(2, 1, 0) = (2b, b, 0)$, $c(1, -1, -1) = (c, -c, -c)$.
Sum: $(a + 2b + c, 2a + b - c, a - c)$.
3. Set each coordinate to 0: Eq. 1: $a + 2b + c = 0$; Eq. 2: $2a + b - c = 0$; Eq. 3: $a - c = 0$.
4. Eq. 3 gives $c = a$.
5. Substitute $c = a$ into Eq. 1: $a + 2b + a = 2a + 2b = 0$, so $b = -a$.
6. Substitute $b = -a$ and $c = a$ into Eq. 2: $2a + (-a) - a = 0$, true for every a . So Eq. 2 adds no new restriction and a is a free variable.
7. Choose $a = 1$ (any nonzero value): $(a, b, c) = (1, -1, 1)$, a nontrivial solution.
8. Check: $v_1 - v_2 + v_3 = (1, 2, 1) - (2, 1, 0) + (1, -1, -1) = (1 - 2 + 1, 2 - 1 - 1, 1 - 0 - 1) = (0, 0, 0)$
✓.

Answer: Linearly dependent: $v_1 - v_2 + v_3 = \mathbf{0}$.

Example 5 · Independence in P_2

Show $\{1 + x, x + x^2, 1 + x^2\}$ is linearly independent in $P_2(\mathbb{R})$.

Solution.

1. Suppose $a(1 + x) + b(x + x^2) + c(1 + x^2)$ is the zero polynomial.
2. Expand and collect powers of x : $a + ax + bx + bx^2 + c + cx^2 = (a + c) + (a + b)x + (b + c)x^2$.
3. A polynomial is zero only if all coefficients are 0. Constant term: $a + c = 0$. Coefficient of x : $a + b = 0$. Coefficient of x^2 : $b + c = 0$.
4. From $a + c = 0$: $a = -c$. From $a + b = 0$: $b = -a = c$.
5. Substitute into $b + c = 0$: $c + c = 2c = 0$, so $c = 0$.
6. Then $a = -c = 0$ and $b = c = 0$. Only the trivial solution exists, so the set is linearly independent.

Answer: Linearly independent.

Example 6 · The field matters

Is $\{(1, i), (i, -1)\}$ linearly independent in $\mathbb{C}^2$ over $\mathbb{C}$? Over $\mathbb{R}$?

Solution.

1. Write $v_1 = (1, i)$ and $v_2 = (i, -1)$. First look for a complex multiple: $i \cdot v_1 = (i \cdot 1, i \cdot i) = (i, i^2) = (i, -1) = v_2$ (using $i^2 = -1$).
2. Over $\mathbb{C}$: $i v_1 - 1 \cdot v_2 = \mathbf{0}$ is a relation with coefficients i and -1 , not both zero. So the set is dependent over $\mathbb{C}$.
3. Over $\mathbb{R}$: suppose $a(1, i) + b(i, -1) = (0, 0)$ with real a, b . Expand: $(a + bi, ai - b) = (0, 0)$.
4. First coordinate $a + bi = 0$: a complex number is 0 only when both real and imaginary parts are 0, so $a = 0$ and $b = 0$.
5. Only the trivial solution exists over $\mathbb{R}$, so the set is independent over $\mathbb{R}$.

Answer: Dependent over $\mathbb{C}$; independent over $\mathbb{R}$.

Example 7 · Proof: dependence means one vector is redundant

Prove: if $v_1, \dots, v_k$ are linearly dependent, then some v_j is a linear combination of the others.

Solution.

1. By the definition of dependence there are scalars $a_1, \dots, a_k$, *not all zero*, with $a_1 v_1 + \dots + a_k v_k = \mathbf{0}$.
2. Since not all the a_i are zero, pick an index j with $a_j \neq 0$.
3. Isolate the j -th term: move all other terms to the right side, $a_j v_j = -\sum_{i \neq j} a_i v_i$.
4. Divide both sides by a_j . This is legal because $a_j \neq 0$ and every nonzero element of a field has an inverse: $v_j = -\sum_{i \neq j} \frac{a_i}{a_j} v_i$.
5. The right side is a linear combination of the vectors v_i with $i \neq j$, so v_j lies in their span.

Answer: $v_j \in \text{span}\{v_i : i \neq j\}$.

Example 8 · Describing a span by an equation

Find the condition on (a, b, c) for $(a, b, c) \in \text{span}\{(1, 1, 0), (0, 1, 1)\}$.

Solution.

1. We need numbers x, y with $x(1, 1, 0) + y(0, 1, 1) = (a, b, c)$.
2. Expand: $x(1, 1, 0) = (x, x, 0)$ and $y(0, 1, 1) = (0, y, y)$. Add: $(x, x + y, y)$.
3. Match coordinates: $x = a, x + y = b, y = c$.
4. The first and third equations give $x = a$ and $y = c$.
5. Substitute into the middle equation: $a + c = b$, i.e. $a - b + c = 0$.
6. Converse: if $a - b + c = 0$ then $b = a + c$ and $a(1, 1, 0) + c(0, 1, 1) = (a, a + c, c) = (a, b, c)$. So the condition is exactly right.

Answer: $a - b + c = 0$

Practice Problems

Part A — Linear Combinations and Span

1.

[Foundational]

Write $(5, -1)$ as a linear combination of $(1, 1)$ and $(1, -1)$ in $\mathbb{R}^2$.

Step 1. Meaning of the question: a *linear combination* of two vectors is a sum of scalar multiples of them. So we must find numbers a and b with $a(1, 1) + b(1, -1) = (5, -1)$.

Step 2. Multiply out the left side. Scalar times a vector multiplies each coordinate: $a(1, 1) = (a, a)$ and $b(1, -1) = (b, -b)$.

Step 3. Add the two vectors coordinate by coordinate: $(a, a) + (b, -b) = (a + b, a - b)$.

Step 4. Two vectors are equal exactly when their coordinates are equal. So $(a + b, a - b) = (5, -1)$ gives two equations: $a + b = 5$ and $a - b = -1$.

Step 5. Solve by adding the two equations (the b terms cancel): $(a + b) + (a - b) = 5 + (-1)$, so $2a = 4$ and therefore $a = 2$.

Step 6. Put $a = 2$ into $a + b = 5$: $2 + b = 5$, so $b = 3$.

Step 7. Check by rebuilding the vector: $2(1, 1) + 3(1, -1) = (2, 2) + (3, -3) = (2 + 3, 2 - 3) = (5, -1)$. This is the target, so the answer is correct.

Answer: $(5, -1) = 2(1, 1) + 3(1, -1)$

2.

[Foundational]

Decide whether $(2, 3, 4) \in \text{span}\{(1, 1, 1), (0, 1, 2)\}$ in $\mathbb{R}^3$. If so, give the coefficients.

Step 1. Meaning: the *span* of two vectors is the set of all their linear combinations. So $(2, 3, 4)$ is in the span exactly when there are numbers a, b with $a(1, 1, 1) + b(0, 1, 2) = (2, 3, 4)$.

Step 2. Compute the left side: $a(1, 1, 1) = (a, a, a)$ and $b(0, 1, 2) = (0, b, 2b)$. Adding coordinatewise: $(a, a + b, a + 2b)$.

Step 3. Match the three coordinates with $(2, 3, 4)$. This gives three equations: first coordinate $a = 2$; second coordinate $a + b = 3$; third coordinate $a + 2b = 4$.

Step 4. The first equation says directly $a = 2$.

Step 5. Put $a = 2$ into the second equation: $2 + b = 3$, so $b = 1$.

Step 6. We have two unknowns but three equations, so the third equation must be *checked*, not assumed. Substitute $a = 2, b = 1$: $a + 2b = 2 + 2 = 4$. It equals the required 4, so all three equations hold.

Step 7. Conclusion: the numbers $a = 2, b = 1$ work, so the vector is in the span. Verify: $2(1, 1, 1) + 1(0, 1, 2) = (2, 2, 2) + (0, 1, 2) = (2, 3, 4)$.

Answer: Yes: $(2, 3, 4) = 2(1, 1, 1) + 1(0, 1, 2)$.

3.

[Foundational]

Decide whether $(1, 0, 0) \in \text{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

Step 1. We ask whether numbers x, y exist with $x(1, 1, 0) + y(0, 1, 1) = (1, 0, 0)$.

Step 2. Expand the left side: $x(1, 1, 0) = (x, x, 0)$ and $y(0, 1, 1) = (0, y, y)$. Their sum is $(x, x + y, y)$.

Step 3. Match coordinates with $(1, 0, 0)$: first $x = 1$; second $x + y = 0$; third $y = 0$.

Step 4. The first equation gives $x = 1$. The third gives $y = 0$.

Step 5. Now test the second equation with these values: $x + y = 1 + 0 = 1$, but the equation demands $x + y = 0$. Since $1 \neq 0$ the three equations contradict each other: no pair (x, y) satisfies all three.

Step 6. Because no such numbers exist, $(1, 0, 0)$ cannot be written as a linear combination of the two vectors.

Answer: No, $(1, 0, 0) \notin \text{span}\{(1, 1, 0), (0, 1, 1)\}$.

4.

[Intermediate]

Find k such that $(1, 2, k) \in \text{span}\{(1, 1, 0), (1, -1, 2)\}$.

Step 1. We need numbers a, b with $a(1, 1, 0) + b(1, -1, 2) = (1, 2, k)$, and then read off what k must be.

Step 2. Expand: $a(1, 1, 0) = (a, a, 0)$ and $b(1, -1, 2) = (b, -b, 2b)$. Adding gives $(a + b, a - b, 2b)$.

Step 3. Match coordinates with $(1, 2, k)$: first $a + b = 1$; second $a - b = 2$; third $2b = k$.

Step 4. The first two equations contain only a and b , so solve them first. Add them: $(a + b) + (a - b) = 1 + 2$, so $2a = 3$ and $a = \frac{3}{2}$.

Step 5. Put $a = \frac{3}{2}$ into $a + b = 1$: $\frac{3}{2} + b = 1$, so $b = 1 - \frac{3}{2} = -\frac{1}{2}$.

Step 6. Now the third equation tells us k : $k = 2b = 2(-\frac{1}{2}) = -1$.

Step 7. Check: $\frac{3}{2}(1, 1, 0) - \frac{1}{2}(1, -1, 2) = (\frac{3}{2}, \frac{3}{2}, 0) + (-\frac{1}{2}, \frac{1}{2}, -1) = (1, 2, -1)$. This matches $(1, 2, k)$ with $k = -1$.

Answer: $k = -1$

5.

[Intermediate]

In $\mathbb{C}^2$ over $\mathbb{C}$, write $(1 + i, 2)$ as a linear combination of $(1, i)$ and $(1, -i)$.

Step 1. Here the scalars may be complex numbers. We need $a, b \in \mathbb{C}$ with $a(1, i) + b(1, -i) = (1 + i, 2)$.

Step 2. Expand the left side: $a(1, i) = (a, ai)$ and $b(1, -i) = (b, -bi)$. Sum: $(a + b, ai - bi)$.

Step 3. Match coordinates: $a + b = 1 + i$ (Equation 1) and $ai - bi = 2$, i.e. $i(a - b) = 2$ (Equation 2).

Step 4. Solve Equation 2 for $a - b$ by dividing by i . Recall $\frac{1}{i} = \frac{1}{i} \cdot \frac{-i}{-i} = \frac{-i}{-i^2} = \frac{-i}{1} = -i$. So $a - b = \frac{2}{i} = 2(-i) = -2i$.

Step 5. Add Equation 1 and $a - b = -2i$: $(a + b) + (a - b) = (1 + i) + (-2i)$, so $2a = 1 - i$ and $a = \frac{1 - i}{2}$.

Step 6. Subtract $a - b = -2i$ from Equation 1: $(a + b) - (a - b) = (1 + i) - (-2i)$, so $2b = 1 + 3i$ and $b = \frac{1 + 3i}{2}$.

Step 7. Check Equation 1: $a + b = \frac{1 - i}{2} + \frac{1 + 3i}{2} = \frac{2 + 2i}{2} = 1 + i \checkmark$.

Step 8. Check Equation 2: $a - b = \frac{1 - i}{2} - \frac{1 + 3i}{2} = \frac{-4i}{2} = -2i$, and $i(-2i) = -2i^2 = 2 \checkmark$.

Answer: $a = \frac{1 - i}{2}$, $b = \frac{1 + 3i}{2}$

6.

[Foundational]

Show that $\text{span}\{(1, 2), (2, 4)\} \subseteq \mathbb{R}^2$ is a line, and give its equation.

Step 1. A general element of the span is $a(1, 2) + b(2, 4)$ with a, b any real numbers.

Step 2. Expand: $a(1, 2) = (a, 2a)$ and $b(2, 4) = (2b, 4b)$. The sum is $(a + 2b, 2a + 4b)$.

Step 3. Factor the second coordinate: $2a + 4b = 2(a + 2b)$. So the element equals $(a + 2b, 2(a + 2b))$.

Step 4. Give the repeated quantity a name: let $t = a + 2b$. Then the element is $(t, 2t) = t(1, 2)$.

Step 5. As a and b range over all real numbers, $t = a + 2b$ takes every real value (for example choose $b = 0, a = t$). So the span is exactly $\{(t, 2t) : t \in \mathbb{R}\}$.

Step 6. The points $(x, y) = (t, 2t)$ satisfy $y = 2x$, and every point with $y = 2x$ has this form. So the span is the line $y = 2x$, which passes through the origin.

Answer: The line $y = 2x$ through the origin.

7.

[Intermediate]

Describe $\text{span}\{(1, 0, 2), (0, 1, -1)\} \subseteq \mathbb{R}^3$ by a single linear equation in x, y, z .

Step 1. A general element is $a(1, 0, 2) + b(0, 1, -1)$ with $a, b \in \mathbb{R}$.

Step 2. Expand: $a(1, 0, 2) = (a, 0, 2a)$ and $b(0, 1, -1) = (0, b, -b)$. Sum: $(a, b, 2a - b)$.

Step 3. Call the coordinates of this vector (x, y, z) . Then $x = a, y = b, z = 2a - b$.

Step 4. Eliminate a and b : replace a by x and b by y in the formula for z : $z = 2x - y$.

Step 5. Move everything to one side to get a standard equation: $2x - y - z = 0$.

Step 6. We must also confirm the equation describes *exactly* the span (no more, no less). Every vector in the span satisfies it, as just shown. Conversely, if (x, y, z) satisfies $z = 2x - y$, then $(x, y, z) = (x, y, 2x - y) = x(1, 0, 2) + y(0, 1, -1)$ (check: $(x, 0, 2x) + (0, y, -y) = (x, y, 2x - y)$), so it lies in the span.

Answer: $\{(x, y, z) : 2x - y - z = 0\}$

8.

[Challenge]

Prove that for any vectors $v_1, \dots, v_k$ in a vector space V , the set $\text{span}\{v_1, \dots, v_k\}$ is a subspace of V .

Step 1. Recall the subspace test: a subset W of V is a subspace if (i) $\mathbf{0} \in W$, (ii) $u, w \in W \Rightarrow u + w \in W$, and (iii) $\lambda \in \mathbb{F}, w \in W \Rightarrow \lambda w \in W$. Let $W = \text{span}\{v_1, \dots, v_k\}$, the set of all $a_1v_1 + \dots + a_kv_k$ with $a_i \in \mathbb{F}$.

Step 2. (i) Zero vector. Take every coefficient equal to 0: $0v_1 + \dots + 0v_k = \mathbf{0} + \dots + \mathbf{0} = \mathbf{0}$, using the fact that $0 \cdot v = \mathbf{0}$ in any vector space. So $\mathbf{0} \in W$.

Step 3. (ii) Closed under addition. Take two elements $u = \sum a_i v_i$ and $w = \sum b_i v_i$ of W . Then $u + w = \sum a_i v_i + \sum b_i v_i$. Using commutativity and associativity of addition, group the terms with the same v_i , and use distributivity $a_i v_i + b_i v_i = (a_i + b_i)v_i$: $u + w = \sum (a_i + b_i)v_i$. This is a linear combination of $v_1, \dots, v_k$ (the new coefficients $a_i + b_i$ are in $\mathbb{F}$), so $u + w \in W$.

Step 4. (iii) Closed under scalar multiplication. Take $\lambda \in \mathbb{F}$ and $u = \sum a_i v_i \in W$. Distributivity gives $\lambda u = \sum \lambda(a_i v_i)$ and the axiom $\lambda(av) = (\lambda a)v$ gives $\lambda u = \sum (\lambda a_i)v_i$, again a linear combination, so $\lambda u \in W$.

Step 5. All three conditions of the subspace test hold, so W is a subspace of V .

Answer: $\text{span}\{v_1, \dots, v_k\}$ is a subspace of V .

Part B — Linear Independence and Dependence

9.

[Foundational]

 Is $\{(1, 2), (3, 6)\}$ linearly independent in $\mathbb{R}^2$?

Step 1. Definition: vectors v_1, v_2 are *linearly dependent* if some combination $av_1 + bv_2 = \mathbf{0}$ holds with a, b not both zero. If the only solution is $a = b = 0$ they are *independent*.

Step 2. Look for a visible relationship. Compare the coordinates of $(3, 6)$ with those of $(1, 2)$: $3 = 3 \cdot 1$ and $6 = 3 \cdot 2$. So $(3, 6) = 3(1, 2)$.

Step 3. Move everything to one side: $3(1, 2) - 1 \cdot (3, 6) = (3, 6) - (3, 6) = (0, 0) = \mathbf{0}$.

Step 4. This is a combination equal to $\mathbf{0}$ with coefficients 3 and -1 , which are not both zero. So a nontrivial relation exists and the set is dependent.

Answer: Linearly dependent.

10.

[Foundational]

 Is $\{(1, 2), (2, 3)\}$ linearly independent in $\mathbb{R}^2$?

Step 1. Set up the test: suppose $a(1, 2) + b(2, 3) = \mathbf{0} = (0, 0)$. We must decide whether $a = b = 0$ is the *only* solution.

Step 2. Expand: $a(1, 2) = (a, 2a)$ and $b(2, 3) = (2b, 3b)$. The sum is $(a + 2b, 2a + 3b)$.

Step 3. Set each coordinate equal to 0: $a + 2b = 0$ (Eq. 1) and $2a + 3b = 0$ (Eq. 2).

Step 4. From Eq. 1, $a = -2b$.

Step 5. Substitute this into Eq. 2: $2(-2b) + 3b = -4b + 3b = -b$. So Eq. 2 becomes $-b = 0$, giving $b = 0$.

Step 6. Then $a = -2b = -2(0) = 0$.

Step 7. The only solution is $a = b = 0$, the trivial one. By the definition, the vectors are linearly independent. (Shortcut that agrees: neither vector is a scalar multiple of the other.)

Answer: Linearly independent.

11.

[Intermediate]

 Is $\{(1, 0, 1), (1, 1, 0), (0, 1, 1)\}$ linearly independent in $\mathbb{R}^3$?

Step 1. Suppose $a(1, 0, 1) + b(1, 1, 0) + c(0, 1, 1) = (0, 0, 0)$. We test whether $a = b = c = 0$ is forced.

Step 2. Expand each term: $a(1, 0, 1) = (a, 0, a)$, $b(1, 1, 0) = (b, b, 0)$, $c(0, 1, 1) = (0, c, c)$. Add coordinatewise: $(a + b, b + c, a + c)$.

Step 3. Set each coordinate to 0: $a + b = 0$ (Eq. 1), $b + c = 0$ (Eq. 2), $a + c = 0$ (Eq. 3).

Step 4. From Eq. 1: $a = -b$. From Eq. 2: $c = -b$.

Step 5. Substitute both into Eq. 3: $a + c = (-b) + (-b) = -2b$, so Eq. 3 becomes $-2b = 0$, hence $b = 0$.

Step 6. Then $a = -b = 0$ and $c = -b = 0$.

Step 7. Only the trivial solution $a = b = c = 0$ exists, so the three vectors are linearly independent.

Answer: Linearly independent.

12.

[Intermediate]

 Is $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ linearly independent in $\mathbb{R}^3$? If not, give a nontrivial relation.

Step 1. Call the vectors $v_1 = (1, 2, 3)$, $v_2 = (4, 5, 6)$, $v_3 = (7, 8, 9)$ and suppose $av_1 + bv_2 + cv_3 = \mathbf{0}$.

Step 2. Expand coordinatewise: $(a + 4b + 7c, 2a + 5b + 8c, 3a + 6b + 9c) = (0, 0, 0)$. Three equations: Eq. 1: $a + 4b + 7c = 0$; Eq. 2: $2a + 5b + 8c = 0$; Eq. 3: $3a + 6b + 9c = 0$.

Step 3. Subtract Eq. 1 from Eq. 2 (eliminating the pattern): $(2a - a) + (5b - 4b) + (8c - 7c) = 0$, i.e. $a + b + c = 0$.

Step 4. Subtract Eq. 2 from Eq. 3: $(3a - 2a) + (6b - 5b) + (9c - 8c) = 0$, again $a + b + c = 0$. The two differences gave the *same* equation, so we have only two independent equations for three unknowns; a free variable will appear.

Step 5. Use Eq. 1 and subtract the equation $a + b + c = 0$ from it: $(a + 4b + 7c) - (a + b + c) = 0$, so $3b + 6c = 0$, which gives $b = -2c$.

Step 6. Put $b = -2c$ into $a + b + c = 0$: $a - 2c + c = 0$, so $a = c$.

Step 7. Choose the free variable $c = 1$ (any nonzero value works). Then $b = -2$ and $a = 1$: the relation is $1 \cdot v_1 - 2 \cdot v_2 + 1 \cdot v_3 = \mathbf{0}$.

Step 8. Check: $v_1 - 2v_2 + v_3 = (1, 2, 3) - (8, 10, 12) + (7, 8, 9) = (1 - 8 + 7, 2 - 10 + 8, 3 - 12 + 9) = (0, 0, 0)$ ✓. A nontrivial relation exists, so the set is dependent.

Answer: Linearly dependent: $v_1 - 2v_2 + v_3 = \mathbf{0}$.

13.

[Challenge]

 Find all real k for which $\{(1, 1, 1), (1, 2, 3), (1, 3, k)\}$ is linearly dependent.

Step 1. Suppose $a(1, 1, 1) + b(1, 2, 3) + c(1, 3, k) = \mathbf{0}$. The set is dependent exactly when this system has a solution other than $a = b = c = 0$.

Step 2. Expand coordinatewise: Eq. 1: $a + b + c = 0$; Eq. 2: $a + 2b + 3c = 0$; Eq. 3: $a + 3b + kc = 0$.

Step 3. Subtract Eq. 1 from Eq. 2: $(a + 2b + 3c) - (a + b + c) = b + 2c = 0$. Call this Eq. 4: $b + 2c = 0$.

Step 4. Subtract Eq. 2 from Eq. 3: $(a + 3b + kc) - (a + 2b + 3c) = b + (k - 3)c = 0$. Call this Eq. 5: $b + (k - 3)c = 0$.

Step 5. Subtract Eq. 4 from Eq. 5: $[b + (k - 3)c] - [b + 2c] = (k - 3 - 2)c = (k - 5)c = 0$.

Step 6. Case $k \neq 5$: then $k - 5 \neq 0$, so $c = 0$. Eq. 4 gives $b = -2c = 0$ and Eq. 1 gives $a = -b - c = 0$. Only the trivial solution: independent.

Step 7. Case $k = 5$: the equation $(k - 5)c = 0$ becomes $0 = 0$ and puts no restriction on c , so c is free. Take $c = 1$: Eq. 4 gives $b = -2$, Eq. 1 gives $a = -b - c = 2 - 1 = 1$.

Step 8. Check for $k = 5$: $1(1, 1, 1) - 2(1, 2, 3) + 1(1, 3, 5) = (1 - 2 + 1, 1 - 4 + 3, 1 - 6 + 5) = (0, 0, 0)$ ✓. So at $k = 5$ there is a nontrivial relation.

Answer: Dependent exactly when $k = 5$.

14.

[Foundational]

 In $P_2(\mathbb{R})$, is $\{1 + x, 1 - x, 2\}$ linearly independent?

Step 1. $P_2(\mathbb{R})$ is the space of real polynomials of degree at most 2; the “vectors” are polynomials, and $\mathbf{0}$ is the zero polynomial.

Step 2. Look for a visible relationship by adding the first two polynomials: $(1 + x) + (1 - x) = 1 + x + 1 - x = 2$.

Step 3. So $(1 + x) + (1 - x) = 2$. Move 2 to the left: $1 \cdot (1 + x) + 1 \cdot (1 - x) + (-1) \cdot 2 = 0$.

Step 4. This is a combination equal to the zero polynomial with coefficients $1, 1, -1$, not all zero. So the set is dependent.

Answer: Linearly dependent.

15.

[Intermediate]

In $\mathbb{C}^2$, consider $\{(1, i), (1+i, i-1)\}$. (a) Is it linearly independent over $\mathbb{C}$? (b) Is it linearly independent over $\mathbb{R}$?

Step 1. Write $v_1 = (1, i)$ and $v_2 = (1+i, i-1)$. The answer depends on which scalars are allowed: complex scalars (part a) or real scalars only (part b).

Step 2. (a) Try to find a complex multiple. Multiply v_1 by $1+i$: $(1+i)(1, i) = (1+i, (1+i)i)$. Compute $(1+i)i = i + i^2 = i - 1$. So $(1+i)v_1 = (1+i, i-1) = v_2$.

Step 3. Hence $(1+i)v_1 - 1 \cdot v_2 = \mathbf{0}$, a relation with coefficients $1+i$ and -1 (not both zero). Over $\mathbb{C}$ the set is dependent.

Step 4. (b) Now allow only real a, b . Suppose $a(1, i) + b(1+i, i-1) = (0, 0)$ with $a, b \in \mathbb{R}$.

Step 5. First coordinate: $a \cdot 1 + b(1+i) = a + b + bi = 0$. A complex number is zero only if its real and imaginary parts are both zero. Real part: $a + b = 0$. Imaginary part: $b = 0$.

Step 6. From $b = 0$ and $a + b = 0$ we get $a = 0$. (We should also confirm the second coordinate does not change this: with $a = b = 0$ it is satisfied.)

Step 7. Over $\mathbb{R}$ only the trivial relation exists, so the set is independent. The same two vectors are dependent over $\mathbb{C}$ but independent over $\mathbb{R}$: independence depends on the field of scalars.

Answer: (a) dependent over $\mathbb{C}$; (b) independent over $\mathbb{R}$.

16.

[Challenge]

Let u, v, w be linearly independent in a real vector space. Prove that $u + v$, $v + w$, $u + w$ are linearly independent.

Step 1. To prove independence we use the definition: assume a combination of the three new vectors is $\mathbf{0}$ and show all its coefficients must be 0. Suppose $a(u + v) + b(v + w) + c(u + w) = \mathbf{0}$ with $a, b, c \in \mathbb{R}$.

Step 2. Distribute the scalars: $au + av + bv + bw + cu + cw = \mathbf{0}$.

Step 3. Collect the terms containing each of u, v, w (commutativity of addition and distributivity): $(a + c)u + (a + b)v + (b + c)w = \mathbf{0}$.

Step 4. Since u, v, w are linearly independent, a combination of them equals $\mathbf{0}$ only if every coefficient is 0. Therefore $a + c = 0$, $a + b = 0$, $b + c = 0$.

Step 5. Add the three equations: $2a + 2b + 2c = 0$, so $a + b + c = 0$.

Step 6. Subtract each original equation from $a + b + c = 0$: $a = (a + b + c) - (b + c) = 0$; $b = (a + b + c) - (a + c) = 0$; $c = (a + b + c) - (a + b) = 0$.

Step 7. We started from a vanishing combination and proved all coefficients a, b, c are 0. Hence $u + v$, $v + w$, $u + w$ are linearly independent.

Answer: Only the trivial relation exists, so they are linearly independent.

Part C — Setting Up Systems, Spanning Sets and Proofs

17.

[Foundational]

 Set up and solve the system that decides whether $(1, 2, 3) \in \text{span}\{(1, 0, 1), (2, 1, 0), (0, 1, 1)\}$.

Step 1. We need numbers a, b, c with $a(1, 0, 1) + b(2, 1, 0) + c(0, 1, 1) = (1, 2, 3)$.

Step 2. Expand each term: $a(1, 0, 1) = (a, 0, a)$, $b(2, 1, 0) = (2b, b, 0)$, $c(0, 1, 1) = (0, c, c)$. Add: $(a + 2b, b + c, a + c)$.

Step 3. Match the coordinates of $(1, 2, 3)$: Eq. 1: $a + 2b = 1$; Eq. 2: $b + c = 2$; Eq. 3: $a + c = 3$.

Step 4. The same system as an augmented matrix (one column per vector, then a bar, then the target):

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 \end{array} \right).$$
Step 5. Solve by substitution. From Eq. 1: $a = 1 - 2b$. From Eq. 2: $c = 2 - b$.

Step 6. Substitute both into Eq. 3: $(1 - 2b) + (2 - b) = 3$, i.e. $3 - 3b = 3$, so $-3b = 0$ and $b = 0$.

Step 7. Back-substitute: $a = 1 - 2(0) = 1$ and $c = 2 - 0 = 2$.

Step 8. Check: $1(1, 0, 1) + 0(2, 1, 0) + 2(0, 1, 1) = (1, 0, 1) + (0, 0, 0) + (0, 2, 2) = (1, 2, 3)$ ✓. A solution exists, so the vector is in the span.

Answer: Yes: $(1, 2, 3) = 1(1, 0, 1) + 0(2, 1, 0) + 2(0, 1, 1)$.

18.

[Intermediate]

 Find the condition on (a, b, c) for $(a, b, c) \in \text{span}\{(1, 2, 0), (0, 1, 1)\}$.

Step 1. A vector (a, b, c) is in the span exactly when there are numbers x, y with $x(1, 2, 0) + y(0, 1, 1) = (a, b, c)$. (We use x, y for the coefficients because a, b, c are already the target's coordinates.)

Step 2. Expand: $x(1, 2, 0) = (x, 2x, 0)$ and $y(0, 1, 1) = (0, y, y)$. The sum is $(x, 2x + y, y)$.

Step 3. Match coordinates: $x = a$, $2x + y = b$, $y = c$.

Step 4. The first and third equations give the coefficients directly: $x = a$ and $y = c$.

Step 5. Substitute these into the middle equation: $2a + c = b$. This is the condition that the target must satisfy for the system to be consistent.

Step 6. Rearrange into standard form: $2a - b + c = 0$.

Step 7. Converse: if $2a - b + c = 0$ then $b = 2a + c$, and $a(1, 2, 0) + c(0, 1, 1) = (a, 2a + c, c) = (a, b, c)$, so such a vector is indeed in the span.

Answer: $2a - b + c = 0$

19.

[Intermediate]

 Find a finite spanning set for $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$.

Step 1. A *spanning set* is a list of vectors whose linear combinations give every element of W . We must describe all solutions of $x + y + z = 0$.

Step 2. Solve the equation for one variable: $x = -y - z$. The variables y and z are free (they can be any real numbers).

Step 3. So every element of W has the form $(x, y, z) = (-y - z, y, z)$.

Step 4. Split this vector according to its free variables: $(-y - z, y, z) = (-y, y, 0) + (-z, 0, z) = y(-1, 1, 0) + z(-1, 0, 1)$.

Step 5. Therefore every element of W is a linear combination of $(-1, 1, 0)$ and $(-1, 0, 1)$: $W \subseteq$

$$\text{span}\{(-1, 1, 0), (-1, 0, 1)\}.$$

Step 6. For the reverse inclusion, check that each spanning vector lies in W : $(-1) + 1 + 0 = 0$ and $(-1) + 0 + 1 = 0$. Since W is a subspace (closed under combinations), every combination of them also lies in W .

Step 7. Both inclusions hold, so the two vectors span W .

Answer: $W = \text{span}\{(-1, 1, 0), (-1, 0, 1)\}$

20.

[Intermediate]

(a) Show that $\{(1, 1), (1, 2), (2, 1)\}$ spans $\mathbb{R}^2$. (b) Is it linearly independent?

Step 1. (a) To show the set spans $\mathbb{R}^2$, take an arbitrary target (x, y) and show it is a combination. It is enough to use only the first two vectors: $a(1, 1) + b(1, 2) = (x, y)$.

Step 2. Expand: $(a + b, a + 2b) = (x, y)$, so $a + b = x$ (Eq. 1) and $a + 2b = y$ (Eq. 2).

Step 3. Subtract Eq. 1 from Eq. 2: $b = y - x$. Substitute in Eq. 1: $a = x - b = x - (y - x) = 2x - y$.

Step 4. So for *every* (x, y) we obtain coefficients $a = 2x - y$, $b = y - x$. Check: $(2x - y)(1, 1) + (y - x)(1, 2) = (2x - y + y - x, 2x - y + 2y - 2x) = (x, y)$ ✓. Hence the set spans $\mathbb{R}^2$.

Step 5. (b) The set has 3 vectors in $\mathbb{R}^2$. Look for a relation: write the third vector in terms of the first two. Solve $(2, 1) = a(1, 1) + b(1, 2)$ using the formulas from part (a) with $(x, y) = (2, 1)$: $a = 2(2) - 1 = 3$ and $b = 1 - 2 = -1$.

Step 6. Check: $3(1, 1) - 1(1, 2) = (3, 3) - (1, 2) = (2, 1)$ ✓.

Step 7. Rearranged: $3(1, 1) - 1(1, 2) - 1(2, 1) = (0, 0)$, a combination equal to $\mathbf{0}$ with coefficients $3, -1, -1$ not all zero. So the set is dependent.

Answer: (a) Yes it spans. (b) No, it is linearly dependent.

21.

[Intermediate]

Show that $\{1, 1 + x, 1 + x + x^2\}$ spans $P_2(\mathbb{R})$.

Step 1. $P_2(\mathbb{R})$ consists of all polynomials $a + bx + cx^2$ with real a, b, c . We must show each of them is a combination of the three given polynomials.

Step 2. Take an arbitrary $p(x) = a + bx + cx^2$ and look for real α, β, γ with $a + bx + cx^2 = \alpha \cdot 1 + \beta(1 + x) + \gamma(1 + x + x^2)$.

Step 3. Expand the right side and collect powers of x : $\alpha + \beta + \beta x + \gamma + \gamma x + \gamma x^2 = (\alpha + \beta + \gamma) + (\beta + \gamma)x + \gamma x^2$.

Step 4. Two polynomials are equal exactly when all coefficients agree: coefficient of x^2 : $\gamma = c$; coefficient of x : $\beta + \gamma = b$; constant term: $\alpha + \beta + \gamma = a$.

Step 5. Solve from the top: $\gamma = c$. Then $\beta = b - \gamma = b - c$. Then $\alpha = a - \beta - \gamma = a - (b - c) - c = a - b$.

Step 6. These formulas give coefficients for every a, b, c . Check: $(a - b) \cdot 1 + (b - c)(1 + x) + c(1 + x + x^2) = (a - b) + (b - c) + (b - c)x + c + cx + cx^2 = a + bx + cx^2$ ✓.

Step 7. Every polynomial in $P_2(\mathbb{R})$ is a linear combination of the three, so they span it.

Answer: $a + bx + cx^2 = (a - b) \cdot 1 + (b - c)(1 + x) + c(1 + x + x^2)$; spans.

22.

[Challenge]

Show that $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$ spans $M_{2 \times 2}(\mathbb{R})$ and is linearly independent.

Step 1. Name the matrices $E_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $E_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $E_3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $E_4 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Take an arbitrary matrix $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ and look for $\alpha, \beta, \gamma, \delta$ with $\alpha E_1 + \beta E_2 + \gamma E_3 + \delta E_4 = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$.

Step 2. Compute the left side entry by entry (scalar times matrix multiplies every entry; matrices add entrywise): $\begin{pmatrix} \alpha + \delta & \beta + \gamma \\ \beta - \gamma & \alpha - \delta \end{pmatrix}$.

Step 3. Match the four entries: $\alpha + \delta = p$ (position 1,1); $\beta + \gamma = q$ (1,2); $\beta - \gamma = r$ (2,1); $\alpha - \delta = s$ (2,2).

Step 4. Solve the pair $\alpha + \delta = p$, $\alpha - \delta = s$: add to get $2\alpha = p + s$, so $\alpha = \frac{p+s}{2}$; subtract to get $2\delta = p - s$, so $\delta = \frac{p-s}{2}$.

Step 5. Solve the pair $\beta + \gamma = q$, $\beta - \gamma = r$: add to get $2\beta = q + r$, so $\beta = \frac{q+r}{2}$; subtract to get $2\gamma = q - r$, so $\gamma = \frac{q-r}{2}$.

Step 6. These coefficients exist for *every* choice of p, q, r, s , so every 2×2 real matrix is a combination of $E_1, \dots, E_4$: the set spans $M_{2 \times 2}(\mathbb{R})$.

Step 7. Independence: suppose $\alpha E_1 + \beta E_2 + \gamma E_3 + \delta E_4$ equals the zero matrix. Then $p = q = r = s = 0$ in the equations above.

Step 8. The formulas give $\alpha = \frac{0+0}{2} = 0$, $\delta = 0$, $\beta = 0$, $\gamma = 0$. Only the trivial combination gives the zero matrix, so the four matrices are linearly independent.

Answer: The set spans $M_{2 \times 2}(\mathbb{R})$ and is linearly independent.

23.

[Challenge]

Prove: if $w \in \text{span}\{v_1, \dots, v_k\}$ then $\text{span}\{v_1, \dots, v_k, w\} = \text{span}\{v_1, \dots, v_k\}$.

Step 1. To prove two sets are equal we prove each is contained in the other. Let $S = \text{span}\{v_1, \dots, v_k\}$ and $T = \text{span}\{v_1, \dots, v_k, w\}$.

Step 2. ($S \subseteq T$) Take any $\sum a_i v_i \in S$. Then $\sum a_i v_i = a_1 v_1 + \dots + a_k v_k + 0 \cdot w$, which is a combination of $v_1, \dots, v_k, w$ (coefficient 0 on w). So $S \subseteq T$.

Step 3. ($T \subseteq S$) Because $w \in S$, we can write $w = \sum c_i v_i$ for some scalars $c_1, \dots, c_k$.

Step 4. Take any element of T : $\sum a_i v_i + bw$ for scalars $a_1, \dots, a_k, b$. Substitute $w = \sum c_i v_i$: $\sum a_i v_i + b \sum c_i v_i$.

Step 5. Distribute b and combine like terms: $\sum a_i v_i + \sum bc_i v_i = \sum (a_i + bc_i) v_i$.

Step 6. This is a combination of $v_1, \dots, v_k$ alone, so it lies in S . Hence $T \subseteq S$.

Step 7. Both inclusions hold, so $S = T$: adding a vector that is already in the span does not enlarge the span.

Answer: The two spans are equal.

24.

[Challenge]

Let $v_1, \dots, v_k$ be linearly independent and $v_{k+1} \notin \text{span}\{v_1, \dots, v_k\}$. Prove that $v_1, \dots, v_{k+1}$ are linearly independent.

Step 1. Use the definition. Suppose $a_1v_1 + \dots + a_kv_k + a_{k+1}v_{k+1} = \mathbf{0}$. We must show every $a_j = 0$.

Step 2. First look at the last coefficient a_{k+1} . We argue by contradiction: assume $a_{k+1} \neq 0$.

Step 3. Then we may solve for v_{k+1} : move the other terms to the right, $a_{k+1}v_{k+1} = -a_1v_1 - \dots - a_kv_k$, and divide by a_{k+1} (allowed because $a_{k+1} \neq 0$ in a field): $v_{k+1} = -\frac{a_1}{a_{k+1}}v_1 - \dots - \frac{a_k}{a_{k+1}}v_k$.

Step 4. The right side is a linear combination of $v_1, \dots, v_k$, so $v_{k+1} \in \text{span}\{v_1, \dots, v_k\}$. This contradicts the hypothesis. Therefore $a_{k+1} = 0$.

Step 5. With $a_{k+1} = 0$ the original equation reduces to $a_1v_1 + \dots + a_kv_k = \mathbf{0}$.

Step 6. Since $v_1, \dots, v_k$ are linearly independent, this forces $a_1 = \dots = a_k = 0$.

Step 7. All coefficients $a_1, \dots, a_{k+1}$ are 0, so $v_1, \dots, v_{k+1}$ are linearly independent.

Answer: All coefficients vanish, so $v_1, \dots, v_{k+1}$ are linearly independent.